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GitHub Link	https://githuAb.com/192425292simats-byte/MLA0403-DEEP-LEARNING

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 2

Mock Interview

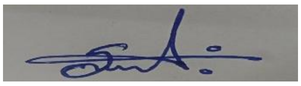
COURSE NAME: Deep Learning **COURSE CODE:MLA0403**
SLOT :A

COURSE OUTCOME COVERED

CO 5: Autoencoders, Undercomplete Autoencoder, Regularized Autoencoder, Sparse Autoencoder, Denoising Autoencoder, Stochastic Encoders and Decoders, Contractive Autoencoder, Deep Belief Networks (DBN), Boltzmann Machines (BM), Deep Boltzmann Machines (DBM), Generative Adversarial Networks (GANs), Generator, Discriminator, GAN Applications.

Course Outcome Weightage: 35% **Duration: 90 minutes**
Assessment Tool Weightage: 20% **Mark : 25**

Code of Conduct:
I (V Sunil Kumar / 192425292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student: **Name of the student:V Sunil**

Criteria	Subject Knowledge and Conceptual Understanding (3)	Technical Accuracy and Application of Concepts (2.5)	Problem-Solving and Analytical Thinking **(2)**	Communication Skills and Clarity of Explanation **(1.5)**	Confidence, Presentation and Professional Behaviour ***(1)***	**Total (10)**
Weightage	***30 %***	***25%***	***20%***	***15 %***	***10 %***	**100%**
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						
Q9						

Q10						
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Q21						
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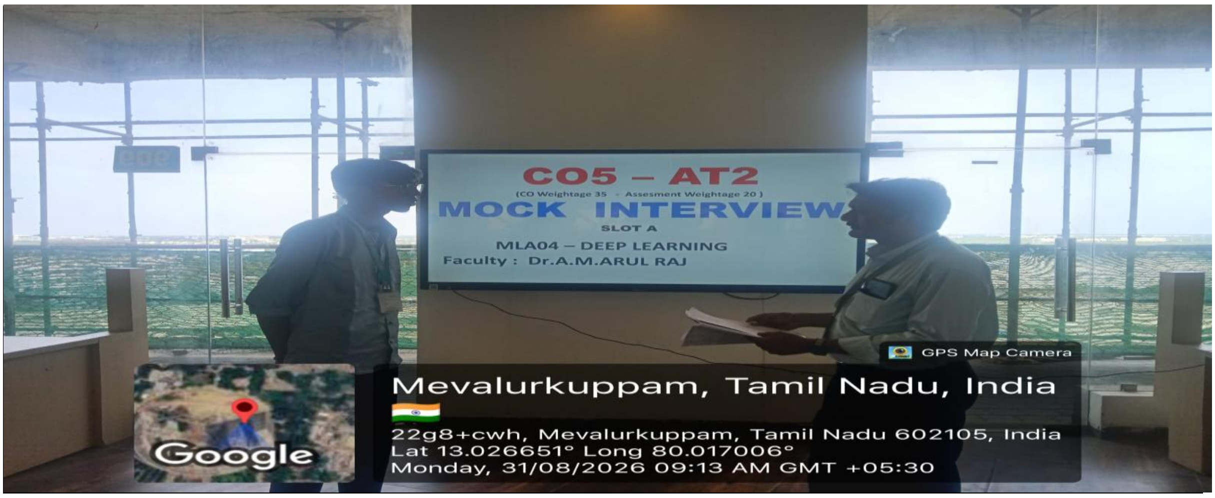
Q23						
Q24						
Q25						

Signature of the Faculty: **Total Marks:**

QUESTIONS: 25 **Total Marks:25**

1. What is an Autoencoder in Deep Learning?
2. What is the main purpose of an Autoencoder?
3. What are the major components of an Autoencoder?
4. What is the role of the Encoder in an Autoencoder?
5. What is the role of the Decoder in an Autoencoder?
6. What is meant by the Bottleneck layer in an Autoencoder?
7. What is meant by Reconstruction in an Autoencoder?
8. How does an Autoencoder learn to reconstruct its input?
9. What is the difference between an Autoencoder and a traditional Neural Network?
10. What type of loss function is commonly used to train an Autoencoder?
11. Explain the basic architecture of an Autoencoder with a suitable example.
12. Why is the dimensionality of the bottleneck layer usually smaller than the input layer?
13. What happens if the bottleneck layer has too many neurons?
14. What happens if the bottleneck layer has very few neurons?
15. What is a Denoising Autoencoder?
16. How can an Autoencoder be used for image denoising?
17. How can Autoencoders be used for dimensionality reduction?
18. How can an Autoencoder be used for anomaly detection?
19. What is the difference between a Denoising Autoencoder and a basic Autoencoder?
20. Give some real-world applications of Autoencoders.
21. What is a Deep Generative Model?
22. What is the difference between a generative model and a discriminative model?
23. What is a Variational Autoencoder (VAE)?
24. What is the difference between a traditional Autoencoder and a Variational Autoencoder (VAE)?
25. What are Generative Adversarial Networks (GANs), and what are the roles of the Generator and Discriminator?

PHOTO :



ANSWERS:

1. **Autoencoder:** An Autoencoder is a neural network that learns to encode input data into a compact representation and reconstruct the original data.
2. **Main purpose:** Its main purpose is to learn useful features and representations of data.
3. **Major components:** The major components are the Encoder, Bottleneck (Latent Space), and Decoder.
4. **Encoder:** The Encoder converts the input data into a lower-dimensional latent representation.
5. **Decoder:** The Decoder reconstructs the original input from the latent representation.
6. **Bottleneck layer:** The Bottleneck is the compressed representation of the input containing its most important features.
7. **Reconstruction:** Reconstruction is the process of generating an output that is as close as possible to the original input.
8. **Learning reconstruction:** An Autoencoder learns by minimizing the difference between the input and reconstructed output using a loss function.
9. **Difference:** An Autoencoder learns to reconstruct its input, while a traditional neural network usually learns to predict a target output.
10. **Loss function:** Mean Squared Error (MSE) is commonly used for continuous data, while Binary Cross-Entropy is often used for normalized binary-like data.
11. **Architecture example:** For an image, the architecture can be **Input → Encoder → Bottleneck → Decoder → Reconstructed Image**.
12. **Smaller bottleneck:** A smaller bottleneck forces the Autoencoder to retain only the most important features of the input.
13. **Too many neurons:** A bottleneck with too many neurons may simply memorize the input

and learn poor feature representations.

14. **Very few neurons:** Too few neurons can cause excessive information loss and poor reconstruction.
15. **Denoising Autoencoder:** A Denoising Autoencoder is trained to reconstruct clean data from a corrupted or noisy input.
16. **Image denoising:** It takes a noisy image as input and learns to produce a cleaner version of the original image.
17. **Dimensionality reduction:** Autoencoders reduce dimensionality by using the compressed representation produced by the bottleneck layer.
18. **Anomaly detection:** Anomalies can be detected when their reconstruction error is significantly higher than that of normal data.
19. **Difference:** A basic Autoencoder reconstructs the original input, while a Denoising Autoencoder reconstructs clean input from intentionally corrupted data.
20. **Applications:** Applications include image denoising, anomaly detection, dimensionality reduction, recommendation systems, and feature extraction.
21. **Deep Generative Model:** A Deep Generative Model is a deep learning model that learns the data distribution and can generate new similar data.
22. **Generative vs discriminative:** Generative models create new data, while discriminative models learn to classify or predict existing data.
23. **VAE:** A Variational Autoencoder is a generative model that learns a probability distribution in the latent space and generates new data from it.
24. **Traditional AE vs VAE:** A traditional Autoencoder learns a fixed latent representation, while a VAE learns a probability distribution in the latent space.
25. **GANs:** GANs consist of a **Generator** that creates fake data and a **Discriminator** that distinguishes real data from fake data.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Subject Knowledge and Conceptual Understanding	30%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Technical Accuracy and Application of Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Problem-Solving and Analytical Thinking	20%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Communication Skills and Clarity of Explanation	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Confidence, Presentation and Professional Behaviour	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis